

National University of Singapore
School of Computing
FINAL EXAMINATION for Semester 2 AY2008/2009

CS1101C — Programming Methodology in C

25 April 2009 Time Allowed: 2 Hours

INSTRUCTIONS TO CANDIDATES

1. This test paper contains 24 questions in two parts A and B, and comprises 14 printed pages, including this page.
2. The maximum possible mark is 40.
3. Answer *all* questions in the *space provided* in this booklet.
4. This is an *open book* examination.
5. Do not look at the questions until you are told to do so.
6. There is no negative marking, so please attempt every question.

Part A (Each question is worth 1 mark. Total: 20 marks)

In this part, you are required to write down the output of the given code fragment, or write down that there is an *error*.

An error includes compiler, runtime or infinite loop error.

Your answers are to be brief and to the point.

Do not include any commentary in the answers.

Assume that the relevant #include pre-processor statements have been included in the code fragment as appropriate.

```
1. main()
{
    int i = 1, i2;
    unsigned int u, u2;
    double d = 2, d2;
    char c = '3', c2, c3;
    i2 = d / 3;
    d2 = (c - '0') / 2;
    c2 = 'a' + d - 0.0001;
    c3 = 'a' + d + 0.0001;
    printf("%d %.1f %c %c\n", i2, d2, c2, c3);
}
```

Handwritten notes:
- $\frac{2}{3}$ for $i2 = d / 3$
- $\frac{2}{3}$ for $d2 = (c - '0') / 2$
- b for $c2$
- c for $c3$
- 0 for $i2$, 1.0 for $d2$
- b for $c2$, c for $c3$
- Red note: "从 double 到 char/int 的强制转换: 舍去小数部分(只保留整数), 且截断方向向零 (truncation towards zero)"

```
2. #define square(x) x*x
main()
{
    int i;
    i = 64/square(4);
    printf("%d", i);
}
```

Handwritten notes:
- Red note: "应写为 ((x)*(x))"
- $64/x^2$ for $i = 64/square(4)$
- $64/4^2$ for $printf$
- 64 for the output

```
3. main()
{
    int i = -1, j = -1, k = 0, l = 2, m;
    m = i++&&j++&&k++||l++;
    printf("%d %d %d %d %d", i, j, k, l, m);
}
```

Handwritten notes:
- -1 for i , 0 for j , 1 for k , 3 for l , 1 for m
- Blue note: "false || 2 $\Rightarrow$ true"
- Blue note: "true && 0 || 2 $\Rightarrow$ true"
- Blue note: "true && 1 || 2 $\Rightarrow$ true"

```

4. main()
{
    int i = 1, j = 2;
    switch (i)
    {
        case 1: printf("good");
                break;
        case (j) printf("bad");
                break;
    }
}

```

good Error

```

5. main()
{
    int i, j = 0;
    for (i = 0; i < 999; i++) {
        switch (i) {
            case 0:
            case 1: if (i < 1) break; i = 555;
            case 2: if (i == 555) j = 666; break;
            default: j = 777;
        }
        if (i == 555) break;
    }
    printf("%d %d\n", i, j);
}

```

555 777 666

```

6. main()
{
    int i = 0;
    do {
        if (i < 3) {
            i += 2; i=2=>i=4
            printf("%d\n", i);
            continue;
        }
        printf("%d\n", ++i);
    } while (i < 3);
}

```

2 4 ✓

```

7. main()
{
    int i = 0;

```

```

do {
    if (i < 3) {
        i += 2;
        printf("%d\n", i);
        break;
    }
    printf("%d\n", ++i);
} while (i < 3);
}

```

8. main()

```

{
    int i, j;
    for (i = 0; i < 3; i++)
        for (j = 0; j <= i; j++)
            printf("%d %d\n", i, j);
}

```

```

0 0
1 0
2 0
2 1
2 2

```

9. int func(int n);

```

main()
{
    printf("%d\n", func(func(21)));
}
int func(int n)
{
    do {
        if ((n-- % 5) == 0) break;
    } while (1);
    return n;
}

```

10. int main()

```

{
    func();
    func();
    return 0;
}

```

void func()

```

{
    int x = 0;
    static int y = 0;
    x++; y++;
    printf( "%d %d\n", x, y );
}

```

11. Describe the output of running the function call fib(4) on the following:

```

int fib(int n) {
    printf("%d ", n);
    return (n <= 1 ? n : fib(n-1) + fib(n-2));
}

```

4 3 2 1 0 1 2 1 0

11. Describe the output of running the function call fib(4) on the following:

```

int fib(int n) {
    printf("%d ", n);
    return (n <= 1 ? n : fib(n-1) + fib(n-2));
}

```

Answer:

4 (fib(3)) + (fib(2))

(3 (fib(2)) + (fib(1)) + ())

((2 (fib(1)) + (fib(0))) + () + ())

(((1) + (0)) + (1) + (210))

4 3 2 1 0 1 2 1 0

4 3 2 1 0 1 2 1 0

12. #define N 100

```
main()
{
    int i, j, count = 0;
    int num[N][N];
    int newnum[N*N];
    for (i = 0; i < N; i++)
        for (j = 0; j < N; j++)
            num[i][j] = i*N + j;
    for (i = 0; i < N; i++)
        for (j = 0; j < N; j++) count = i*N + j
            newnum[count++] = num[i][j];
    printf("%d\n", newnum67[67]); i=0, j=67
    printf("%d\n", newnum0[0]); i=0, j=0
    printf("%d\n", newnum676[876]); i=8, j=6
}
```

67 876

13. main()

```
{
    int i1, i2, *p1, *p2;
    i1 = 5;
    p1 = &i1;
    i2 = *p1 / 2 + 10;
    p2 = p1;
    printf("%d %d %d %d\n", i1, i2, 5*p1, 12*p2); 15 5
}
```

14. main()

```
{
    int a = 2, *f1, *f2;
    f1 = f2 = &a;
    *f2 a=16 += *f2 a=8 += a a=4 += 2;
    printf("%d %d %d\n", a, *f1, 16*f2); 16 16
}
```

赋值运算从右向左计算

```

15. main()
{
    char s[] = "man";
    int i;
    for(i = 0; s[i]; i++)
        printf("%c%c%c\n", s[i], *(s+i), *(i+s));
}

```

Handwritten notes: 'm' 'a' 'n' '\0' (above s[]); m m m / a a a / n n n (next to s[i]); 验证条件: 我们非常, 别非 '\0' (next to s[i]);

```

16. #include <string.h>
main()
{
    char s1[] = "0xyzabc";
    char s2[] = "0xyz";
    char s3[] = "1xyz";
    s1[4] = 0;
    printf("%d %d %d\n", strcmp(s1, s2), strcmp(s2, s3), strcmp(s3, s2));
}

```

Handwritten notes: 0 -> '\0' (left of main()); 10 (above s1[]); 0 < 0 > 0 (below strcmp arguments); 0 -1 1 (below printf);

```

17. #define N 9
void func(int a, int b[], int *c);

main()
{
    int a = 0, b[N] = {0};
    func(a, b, &b[1]);
    printf("%d %d %d %d\n", a, b[0], b[1], b[2]);
}

void func(int a, int b[], int *c)
{
    a = 1;
    b[1] = 2;
    *(c + 1) = 3;
}

```

Handwritten notes: 0 0 2 3 (below printf); b[1] = 2 (underlined);

```

18. main( )
{
    int a[] = {0, 1, 2, 3};
    int *p[] = {a, a+1, a+2, a+3};
    int **ptr = p;
    ptr++;
    printf("%d%d%d\n", ptr-p, *ptr-a, **ptr);
    *ptr++;
    printf("%d%d%d\n", ptr-p, *ptr-a, **ptr);
    **ptr++;
    printf("%d%d%d\n", ptr-p, *ptr-a, **ptr);
}

```

```

19. struct time {
    int *day;
    int *month;
    int *year;
};
struct time t1, *times;

main()
{
    int d = 25, m = 4, y = 2009;

    t1.day = &d;
    t1.month = &m;
    t1.year = &y;
    printf("%d %d %d\n", *t1.day, *t1.month, *t1.year );
    times = &t1;
    *(times->day) = 10;
    printf("%d %d %d\n", *t1.day, *t1.month, *t1.year );
}

```



```
20. main()
{
    int p = 1;
    struct {
        int next;
        char name[10];
    } s[] = {{3, "cow"}, {2, "how"}, {0, "now"}, {-1, "brown"}};

    do {
        printf("%s ", s[p].name);
        p = s[p].next;
    } while (p >= 0);
    printf("\n");
}
```

--

Part B (Each question is worth 5 marks. Total: 20 marks)

In this part, the required code fragment is only a few lines.
Comments are not necessary; if you wish to provide comments
they must be restricted to a few words.

21. We wish to compute the number of months it takes to repay a loan. A fixed monthly payment and an interest rate of one percent *per month* is applicable. For example, given a loan of amount \$100 and a fixed monthly payment of \$40,

After 1 month:

you owe an additional $0.01 * \$100 = \1 , make payment of \$40, so you now owe \$61.

After 2 months:

you owe an additional $0.01 * \$61 = \0.61 , make payment of \$40, so you now owe \$21.61.

After 3 months,

you owe an additional $0.01 * \$21.61 = \0.2161 , make payment of \$40, and now your loan is paid.

(Actually, you have overpaid a bit, by \$18.1739, but we shall ignore this detail.)

So the total time required was 3 months.

Write two versions of a function that computes this loan duration. You may assume that the monthly payment exceeds the value of one month's interest on the loan.

First write a ~~recursive~~ version. *Non-recursive version*

Next write a ~~new~~ recursive version.

--

22. A *magic square* is a square matrix where the sum of the elements in each row, column and diagonal is the same. An example where the number of rows is 3 and the sum in question is 15:

6	1	8
7	5	3
2	9	4

Write a function to determine if a given two dimensional array represents a magic square.

23. Use a structure to represent a *card* whose components are a *rank* from 1 to 13, and a *suit* from 1 to 4. Use an array to represent a *deck* of 52 cards. Write a code fragment to simulate a “shuffling” of the deck. Using the `rand` function, swap the first card with the card at a random position in the rest of the deck, that is, in the second through last positions. Then swap the second card with the card at a random position in the rest of the deck, that is, in the third through last positions. And so on.

Write your code fragment here:

24. We wish to store information about students, modules and grades. Assume that there are *S* students and *M* modules in total.

The following defines a student database: an array *students* of structures each of which represents an individual *student*. A student, in turn, is a structure whose components are a *matricnumber*, a collection of *modules*, and a *count* of the number of modules taken by the student in question. The type of each component is simply integer.

```
#define S ...
#define M ...

typedef struct student {
    int matricnumber;
    int modules[M];
    int count;
} students[S];
```

Define a module database. This should be an array *modules* where each element *module* is a structure with the following components: a *modulecode*, a collection of *students* together with a corresponding *grade*, and a *count* of the number of students having taken the module in question. The type of each component is simply integer. The value of the grade is a number from 0 to 5.

Make your declaration here:

Assume that the databases *students* and *modules* have already been populated with some data. For example, let $S = 2$, $M = 3$ and suppose the two students have matric numbers 111111 and 222222 and the module codes are 1101, 1102 and 1103. Suppose student 111111 took modules 1101, 1103 obtaining grades 3, 4 respectively. Suppose student 222222 took modules 1102, 1103 obtaining grades 1, 5 respectively. Then the average grade or CAP for the students are 3.5 and 3.0 respectively.

Write a code fragment to compute and print the CAP for every student.

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end